Thm $\left\{ \sum_{k\geq 0} \binom{n}{k} a_k b_{n-k} \right\}_{n\geq 0} \longleftrightarrow f(x)g(x)$

Fact $\{a_{n+k}\}_{n\geq 0} \xleftrightarrow{eps} D^k(f(x))$

$\{P(n)\, a_n\}_{n\geq 0} \xleftrightarrow{eps} P(xD) f(x)$

Eg D_n: n-derangement #

i.e. $\left|\{\sigma \in S_n \mid \sigma_i \neq i,\ \forall i \in [n]\}\right|$

$n! - C_n^1 \cdot (n-1)! + C_n^2 \cdot (n-2)! - C_n^3 \cdot (n-3)! \cdots$

$n!\left(1 - 1 + \frac{1}{2} - \frac{1}{6}\right.$

$\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots\right) n!$

Eg C_n n-Catalan #

i.e # of 2n-strings by n "1" and n "0"

# of 0's ≥ # of 1's at any place

Sol: $C_n = \sum C_i C_{n-i}$